UNIVERSITÉ GRENOBLE ALPES

Numerical and experimental study of the thermo-hydro-mechanical behavior of concrete under severe accident conditions using a poro-mechanical approach

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This thesis is submitted in partial fulfillment of the requirements for the Master's degree in the International Program "Geomechanics, Civil Engineering and Risks".

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Finally, I would like to thank my family for their constant encouragement, patience, and support throughout my studies and the completion of this thesis.

Abstract

This thesis investigates the thermo-hydro-mechanical behavior of concrete under severe accident conditions using a poro-mechanical approach.

The study combines numerical modeling and experimental validation to better understand the coupled physics involved.

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Chapter 1

Introduction

1.1 Context and problem statement

Concrete containment structures under severe-accident thermal loading present coupled thermo-hydro-mechanical challenges [2].

1.1.1 High-temperature concrete behaviour in nuclear containment structures

Main contextes:

- Concrete in nuclear containment structures is exposed to extreme temperatures during severe accidents.
- Rapid heating leads to strong thermal stresses, dehydration, cracking and possible spalling govern structural integrity.

1.1.2 Problem statement

Main problems:

- Complex multi-physics coupling under high temperature.
- Temperature- and damage-dependent evolution of transport properties with cracking and stiffness degradation altering the permeability and pore pressure.
- Reliable prediction requires a consistent thermo-hydro-mechanical (THM) framework.

1.1.3 Limitations of existing THM models

Main limitations of existing THM models:

- Lack of fully coupled interactions between the damage state and transport properties.
- Lack of consistent regularization for strain localization.
- Limited probabilistic assessment of parameter uncertainty and sensitivity.

1.2 Objectives and scope of the thesis

Main objectives of the thesis:

- Understanding and Developing a THM model
- Temperature- and damage-dependent evolution of stiffness and transport properties.
- Numerical implementation and numerical validation.
- Probabilistic sensitivity analysis: Evaluating parameter uncertainty and identifying the dominant physical parameters.

The scope of this work is limited to:

- Thermal boundaries limited by severe accident conditions.
- Continuum-scale THM modelling by Cast3M.
- Damage modelling including temperature-dependent stiffness and transport properties.
- Probabilistic sensitivity analysis at the material parameter level.

1.3 Research enviroment and Computational tools

1.3.1 3SR Laboratory

The host institution providing the scientific environment, experimental data, and expertise in structural mechanics and risk analysis.

1.3.2 Finite element code - Cast3M

Cast3M (or Castem) is an open-source finite-element code developed at CEA, which is used to implement advanced constitutive laws and custom multiphysics formulations.

Compared with COMSOL Multiphysics, Cast3m typically offers greater transparency and control for implementing tightly coupled THM while COMSOL offers a user-friendly GUI for many built-in multiphysics interfaces.

1.3.3 Probabilistic analysis tool - Persalys

Persalys is the computational tool dedicated to managing parameter uncertainty in material properties.

It is used to perform the probabilistic sensitivity analysis, generate random sampling (e.g., Monte Carlo simulations), and rank the dominant THM parameters.

Chapter 2

Theoretical and Numerical framework

2.1 Thermal behavior

2.1.1 Conduction, convection and radiation

Heat transfer is the transport of thermal energy driven by a temperature difference. The following definitions of heat-transfer modes are summarized from standard references [1].

Conduction

Conduction may be viewed as the transfer of energy from the more energetic to the less energetic particles of a substance due to interactions between the particles.

- **Physical Mechanism:** In a gas, conduction is associated with the net transfer of energy by random molecular motion, often referred to as a diffusion of energy. In a solid, conduction may be attributed to atomic activity in the form of lattice vibrations, and in a conductor, it is also due to the translational motion of the free electrons.

- **Rate Equation (Fourier's Law):** For a one-dimensional plane wall having a temperature distribution $T(x)$, the rate equation is expressed as Fourier's law:

$$q''_x = -k\frac{dT}{dx} \tag{2.1}$$

  Where q''_x is the heat flux (W/m^2), k is the thermal conductivity $(W/m \cdot K)$, and the minus sign indicates that heat is transferred in the direction of decreasing temperature.

Convection

The convection heat transfer mode is comprised of two mechanisms: energy transfer due to random molecular motion (diffusion) and energy transfer by the bulk, or macroscopic, motion of the fluid.

- **Physical Mechanism:** Convection heat transfer occurs between a fluid in motion and a bounding surface when the two are at different temperatures, which leads to the development of a hydrodynamic (velocity) boundary layer and a thermal boundary layer.
- **Classification:** Convection is classified as forced convection when the flow is caused by external means (like a fan or pump), or free (natural) convection when the flow is induced by buoyancy forces due to density differences. There are also convection processes accompanied by latent heat exchange, such as boiling and condensation [6].
- **Rate Equation (Newton's Law of Cooling):** Regardless of the nature of the convection process, the appropriate rate equation is [7]:

$$q'' = h(T_s - T_\infty) \tag{2.2}$$

  Where q'' is the convective heat flux (W/m^2), h is the convection heat transfer coefficient $(W/m^2 \cdot K)$, T_s is the surface temperature, and T_∞ is the fluid temperature.

Radiation

Thermal radiation is energy emitted by matter that is at a nonzero temperature, and unlike conduction and convection, it does not require the presence of a material medium.

- **Physical Mechanism:** The energy of the radiation field is transported by electromagnetic waves (or photons) and may be attributed to changes in the electron configurations of the constituent atoms or molecules.
- **Emissive Power:** The upper limit to the emissive power (rate of energy released per unit area) is prescribed by the Stefan-Boltzmann law for an ideal radiator or blackbody:

$$E_b = \sigma T_s^4 \tag{2.3}$$

  The heat flux emitted by a real surface is given by $E = \varepsilon\sigma T_s^4$, where σ is the Stefan-Boltzmann constant and ε is the emissivity.
- **Irradiation and Absorption:** The rate at which radiation is incident on a unit area of the surface from its surroundings is called irradiation, G. The rate at which radiant energy is absorbed is given by $G_{abs} = \alpha G$, where α is the absorptivity.

- **Net Radiation Exchange:** For a gray surface ($\alpha = \varepsilon$) at T_s completely surrounded by a much larger isothermal surface at T_{sur}, the net rate of radiation heat transfer per unit area is:
$$q''_{rad} = \varepsilon\sigma(T_s^4 - T_{sur}^4) \tag{2.4}$$
For many applications, this can also be expressed using a linearized radiation heat transfer coefficient h_r as $q_{rad} = h_r A(T_s - T_{sur})$.

2.1.2 Heat transfer

Heat Equation

Based on the principle of energy conservation and Fourier's law [**di_paola_starting_2023**], the general heat diffusion equation (or heat equation) for a medium can be expressed as:

$$\rho c_p \frac{\partial T}{\partial t} + \mathrm{div}(-\lambda \overrightarrow{\mathrm{grad}}(T)) - q = 0 \tag{2.5}$$

Where:

- ρ is the volumetric mass density (kg/m^3).
- c_p is the specific heat capacity ($J/kg \cdot K$).
- λ (or k) is the thermal conductivity ($W/m \cdot K$).
- T is the temperature field and t is time.
- q is the volume heat flux source (W/m^3).

To solve this equation, appropriate boundary conditions must be specified:

- **Prescribed temperatures (Dirichlet condition):** The temperature is specified as $T = T_{imp}$ on the boundary surface ∂V^T.
- **Prescribed surface heat flux (Neumann / Mixed condition):** The total heat flux due to conduction, convection, and radiation on the boundary surface ∂V^φ is given by:

$$\vec{n} \cdot (\lambda \overrightarrow{\mathrm{grad}}(T)) = \varphi_{imp} + h(T_f - T) + \varepsilon\sigma(T_\infty^4 - T^4) \tag{2.6}$$

where φ_{imp} is the prescribed heat flux, $h(T_f - T)$ represents the convection term, and $\varepsilon\sigma(T_\infty^4 - T^4)$ is the radiation term.

Discretization

In many practical cases, analytical solutions to the heat equation are impossible to obtain due to complex geometries or boundary conditions. Instead, the problem is approximated using numerical methods, such as the Finite Element (FE) method.

Using the FE discretization, the temperature and temperature gradient in the spatial domain are expressed in matrix form as:

$$T(x) = [N(x)]\{T\}, \quad \overrightarrow{\text{grad}}(T) = [B(x)]\{T\} \tag{2.7}$$

From the weak formulation of the heat equation combined with the spatial discretization, we obtain the general matrix equation:

$$[C]\{\dot{T}\} + [K]\{T\} = \{F\} \tag{2.8}$$

Where the matrices are defined as follows:

- **Capacity matrix:**
  $[C] = \int_V \rho c_p [N]^T [N] dV \quad (J/K)$
- **Conductivity matrix:**
  $[K] = \int_V [B]^T [\lambda][B] dV + \int_{\partial V^\varphi} h[N]^T [N] dS \quad (W/K)$
- **Equivalent nodal flux vectors:**
  $\{F\} = \int_V [N]^T q dV + \int_{\partial V^\varphi} [N]^T \left(\varphi_{imp} + hT_f + \varepsilon\sigma(T_\infty^4 - T^4)\right) dS \quad (W)$

For steady-state thermal analysis, the temperature is time-independent ($\{\dot{T}\} = 0$). Thus, the system simplifies into a linear algebraic equation (assuming constant material properties and neglecting radiation):

$$[K]\{T\} = \{F\} \tag{2.9}$$

By solving this linear system, we can determine the unknown temperatures $\{T\}$ at the nodes across the entire global mesh.

2.2 Hydric behavior

2.2.1 Moisture transport

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2.2.2 Drying, shrinkage and pore pressure

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2.3 Mechanical behavior

The mechanical response of concrete under high-temperature conditions is characterized by an initial linear elastic phase, followed by severe stiffness degradation due to the formation of micro-cracks. In this numerical framework, the behavior is modeled by coupling a linear elastic constitutive law with a continuum damage mechanics approach.

2.3.1 Elastic model

Kinematics and total strain decomposition

Before any micro-structural degradation occurs, concrete is assumed to behave as an isotropic linear elastic material. Under the small strain hypothesis, the macroscopic total strain tensor $\boldsymbol{\varepsilon}$ is derived from the symmetric part of the displacement gradient vector $\mathbf{u}$:

$$\boldsymbol{\varepsilon} = \frac{1}{2}\left(\nabla\mathbf{u} + \nabla^T\mathbf{u}\right) \tag{2.10}$$

To accurately account for the multi-physics interactions within the Thermo-Hydro-Mechanical (THM) framework, the total strain tensor $\boldsymbol{\varepsilon}$ is additively decomposed into four distinct components:

$$\boldsymbol{\varepsilon} = \boldsymbol{\varepsilon}_e + \boldsymbol{\varepsilon}_{th} + \boldsymbol{\varepsilon}_{rd} + \boldsymbol{\varepsilon}_{re} \tag{2.11}$$

The physical meaning of each component is defined as follows:

- $\boldsymbol{\varepsilon}_e$ is the reversible elastic strain tensor. In the pure elastic regime, it is the only component that actively generates the effective macroscopic Cauchy stress $\boldsymbol{\sigma}$ through

Hooke's law:

$$\boldsymbol{\sigma} = \mathbf{C} : \boldsymbol{\varepsilon}_e = \mathbf{C} : (\boldsymbol{\varepsilon} - \boldsymbol{\varepsilon}_{th} - \boldsymbol{\varepsilon}_{rd} - \boldsymbol{\varepsilon}_{re}) \tag{2.12}$$

where $\mathbf{C}$ is the constant fourth-order linear elastic stiffness tensor.

- $\boldsymbol{\varepsilon}_{th} = \alpha(T - T_{ref})\mathbf{I}$ is the isotropic thermal strain induced by temperature variations, with α being the coefficient of thermal expansion.
- $\boldsymbol{\varepsilon}_{rd} = -K(C_{ref} - C)\mathbf{I}$ is the desiccation shrinkage strain associated with moisture loss (drying).
- $\boldsymbol{\varepsilon}_{re} = -\beta\mu\mathbf{I}$ is the endogenous shrinkage strain related to the hydration process.

Linear finite element formulation

In the context of the Finite Element Method (FEM), the continuous equilibrium equation ($\mathrm{div}\boldsymbol{\sigma} + \mathbf{f} = \mathbf{0}$) is transformed into a discrete weak formulation. The continuous displacement field $\mathbf{u}$ is approximated by the nodal displacements $\mathbf{U}$ using the matrix of shape functions $\mathbf{N}$, and the linear strains are obtained via the derivative matrix $\mathbf{B}$ such that $\boldsymbol{\varepsilon}_{lin} = \mathbf{B}\mathbf{U}$.

Applying the principle of virtual work, the global linear system takes the well-known fundamental form:

$$[\mathbf{K}]\{\mathbf{U}\} = \{\mathbf{F}\} \tag{2.13}$$

where $[\mathbf{K}] = \int_V \mathbf{B}^T\mathbf{C}\mathbf{B}dV$ is the global elastic stiffness matrix, and $\{\mathbf{F}\}$ is the vector of equivalent nodal forces, which accumulates the prescribed surface forces $\{\mathbf{F}\}^S$, reaction forces $\{\mathbf{F}\}^R$, volume forces $\{\mathbf{F}\}^V$, and thermal pseudo-forces. In Cast3M, this linear system is directly assembled and solved using the `RIGI` and `RESO` operators.

2.3.2 Mazars damage model

Concrete cracks when tensile strains become too large. In this work, this behaviour is represented with the Mazars damage model [**hamon_model_2013**].

Damage parameter and equivalent strain

The model introduces a single scalar internal variable, D, ranging from 0 (intact material) to 1 (fully broken material). The apparent macroscopic stress is reduced according to the effective stress concept:

$$\boldsymbol{\sigma} = (1 - D)\mathbf{C} : \boldsymbol{\varepsilon}_e \tag{2.14}$$

Because concrete cracking is primarily governed by tensile extensions, Mazars introduced an *equivalent strain* (ε_{eq}) to translate any 3D multiaxial strain state into a scalar uniaxial indicator. In the case of a thermo-mechanical loading, it is important to note that only the elastic strain tensor $\boldsymbol{\varepsilon}_e$ contributes to the evolution of damage. The equivalent strain is calculated solely from the positive principal elastic strains ε_i:

$$\varepsilon_{eq} = \sqrt{\langle \boldsymbol{\varepsilon}_e \rangle_+ : \langle \boldsymbol{\varepsilon}_e \rangle_+} = \sqrt{\sum_{i=1}^{3} \langle \varepsilon_i \rangle_+^2} \tag{2.15}$$

where the Macaulay brackets $\langle x \rangle_+$ denote the positive part of x (i.e., $\langle x \rangle_+ = x$ if $x \geq 0$, and 0 if $x < 0$).

Damage initiates when the equivalent strain exceeds the initial damage threshold, denoted as ε_{d0}. A simple estimate relates this threshold to the tensile strength f_t and Young's modulus E:

$$\varepsilon_{d0} = \frac{f_t}{E}. \tag{2.16}$$

At the damage onset ($\varepsilon_{eq} = \varepsilon_{d0}$), the damage variable is zero:

$$D = 0 \qquad \text{for} \qquad \varepsilon_{eq} \leq \varepsilon_{d0}. \tag{2.17}$$

For uniaxial tension, damage onset corresponds to:

$$\sigma_t = f_t. \tag{2.18}$$

In this simplified interpretation, f_t denotes the uniaxial tensile strength of the concrete and E is Young's modulus.

Damage evolution laws for tension and compression

Concrete exhibits highly asymmetric behavior, characterized by brittle failure in tension and strain-hardening followed by softening in compression. To accurately represent this, the total damage D is formulated as a combination of two independent damage variables: D_t for tension and D_c for compression.

When the damage threshold is exceeded ($\varepsilon_{eq} > \varepsilon_{d0}$), the evolution of these two modes is explicitly governed by exponential laws:

$$D_t = 1 - \frac{(1 - A_t)\varepsilon_{d0}}{\varepsilon_{eq}} - A_t \exp\left[-B_t(\varepsilon_{eq} - \varepsilon_{d0})\right] \tag{2.19}$$

$$D_c = 1 - \frac{(1 - A_c)\varepsilon_{d0}}{\varepsilon_{eq}} - A_c \exp\left[-B_c(\varepsilon_{eq} - \varepsilon_{d0})\right] \tag{2.20}$$

where A_t, B_t, A_c, and B_c are material parameters identified from uniaxial tensile and compressive tests. These parameters modulate the shape of the post-peak softening curves: parameters A define the residual asymptotic stress level, while parameters B control the slope of the stress drop.

The global macroscopic damage D is then calculated using a linear combination with weighting coefficients α_t and α_c:

$$D = \alpha_t^\beta D_t + \alpha_c^\beta D_c \tag{2.21}$$

The coefficients α_t and α_c depend on the principal stress state, evaluating the respective contributions of tensile and compressive stresses. In pure tension, $\alpha_t = 1$ and $\alpha_c = 0$; conversely, in pure compression, $\alpha_c = 1$ and $\alpha_t = 0$. The parameter β (typically fixed at 1.06) is a shear-correction coefficient introduced to improve the structural response under shear loadings.

Non-linear equilibrium and Imbalance vector (R)

Due to the damage evolution, the stress-strain relationship becomes non-linear. The mechanical equilibrium cannot be solved in a single step like the elastic model. By introducing the non-linear complementary stress $\boldsymbol{\sigma}_{nl}$, the equilibrium is rewritten as:

$$[\mathbf{K}]\{\mathbf{U}\} = \{\mathbf{F}\}_{ext} - \int_V \mathbf{B}^T \boldsymbol{\sigma}_{nl} dV \tag{2.22}$$

To solve this, the `PASAPAS` non-linear procedure evaluates the deviation from equilibrium using an iterative Newton-Raphson-type algorithm. It calculates the *imbalance vector* (or residual), $\{\mathbf{R}\}$, which represents the difference between the external applied forces and the internal resisting forces:

$$\{\mathbf{R}\} = \{\mathbf{F}\}_{ext} - \int_V \mathbf{B}^T \boldsymbol{\sigma} dV \tag{2.23}$$

At each iteration i, the algorithm solves the system using the initial elastic stiffness matrix: $[\mathbf{K}]\{\delta\mathbf{U}\}^{i+1} = \{\mathbf{R}\}^i$. The displacements and internal variables (like D) are continuously updated until the norm of the imbalance vector becomes negligible ($\|\mathbf{R}\| < \epsilon \mathbf{F}_{ref}$).

Mesh dependency and fracture energy regularization

A major drawback of local softening models is the pathological mesh dependency. Mathematically, softening causes a loss of ellipticity in the differential equations, leading to strain localization in a band whose width depends entirely on the finite element size. This results in physically unacceptable ruptures with zero energy dissipation upon mesh refinement.

To overcome this issue without resorting to complex non-local formulations, a regularization technique based on Hillerborg's specific fracture energy (G_f) is utilized. Physically, the fracture energy G_f represents the energy required to propagate a discrete crack per unit *area* (expressed in J/m^2). However, in the continuum damage mechanics framework of the Finite Element Method, cracking is modeled as a smeared softening phenomenon occurring over the *volume* of the 3D finite element.

To reconcile the areal energy dissipation of a real crack with the volumetric dissipation of the continuum element, a characteristic length of the finite element, L_e, is introduced. In the Cast3M implementation (using the `@LONGEF` procedure), this equivalent length is evaluated based on the element's geometry, calculated as $L_e = V^{1/D}$ where D is the spatial dimension.

The total fracture energy G_f dissipated by a finite element of size L_e is the product of L_e and the total area under the uniaxial stress-strain ($\sigma - \varepsilon$) softening curve (which represents the volumetric dissipated energy density $g_f = G_f/L_e$). This total energy is strictly partitioned into two components: the reversible elastic strain energy stored up to the peak stress, and the energy dissipated strictly during the non-linear damage softening phase:

$$G_f = G_{elastic} + G_{softening} \tag{2.24}$$

1. Evaluation of the elastic energy ($G_{elastic}$):
The elastic energy density up to the onset of damage (when the stress reaches the tensile strength f_t at the initial strain threshold $\varepsilon_{d0} = f_t/E_0$) corresponds to the area of the elastic triangle under the curve:

$$g_{elastic} = \frac{1}{2} f_t \varepsilon_{d0} = \frac{f_t^2}{2E_0} \tag{2.25}$$

Multiplying by the characteristic length yields the areal elastic energy: $G_{elastic} = \frac{f_t^2}{2E_0} L_e$. Consequently, the remaining energy strictly available for the damage softening phase is calculated as:

$$G_{softening} = G_f - \frac{f_t^2}{2E_0} L_e \tag{2.26}$$

2. Analytical evaluation of $G_{softening}$ and the softening parameter B_t:
In the original Mazars formulation, the evolution of damage in pure tension is governed by an exponential law. Assuming a purely brittle failure in tension ($A_t \approx 1$), the post-peak stress-strain relationship can be simplified to:

$$\sigma(\varepsilon) \approx f_t \exp\left[-B_t(\varepsilon - \varepsilon_{d0})\right] \tag{2.27}$$

The volumetric softening energy density ($g_{softening}$) is the integral of this stress-strain curve from the peak strain ε_{d0} to infinity:

$$g_{softening} = \int_{\varepsilon_{d0}}^{\infty} f_t \exp\left[-B_t(\varepsilon - \varepsilon_{d0})\right] d\varepsilon = \frac{f_t}{B_t} \tag{2.28}$$

By converting this volumetric density into areal energy via the characteristic length L_e, we establish the fundamental relationship between the fracture energy and the Mazars tensile parameter:

$$G_{softening} = g_{softening} \cdot L_e = \frac{f_t \cdot L_e}{B_t} \tag{2.29}$$

3. Dynamic regularization of B_t (B_t^{mod}):
To ensure that each finite element dissipates exactly the objective intrinsic energy G_f regardless of its size L_e, the parameter B_t must be dynamically recalculated. By equating the integral of the Mazars softening curve to the available softening energy, we deduce the modified parameter B_t^{mod}:

$$B_t^{mod} = \frac{f_t \cdot L_e}{G_{softening}} = \frac{f_t \cdot L_e}{G_f - \frac{f_t^2}{2E_0} L_e} \tag{2.30}$$

In the finite element code Cast3M, this explicit regularization is implemented element-by-element, where the numerator ($f_t \cdot L_e$) and denominator ($G_{softening}$) are defined as `NUM1` and `DEN1`, respectively. This energetic adjustment guarantees that the macroscopic crack propagation and global energy dissipation remain completely objective and independent of the selected spatial discretization.

2.4 THM coupling

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2.5 Damage-dependent property evolution

2.5.1 Young's modulus degradation

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2.5.2 Permeability evolution

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2.5.3 Thermal conductivity and heat capacity evolution

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Chapter 3

Numerical implementation

3.1 Modeling strategy and assumptions

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3.2 Geometry, material properties, boundary conditions and loadings

3.2.1 Geometry

3.2.2 Boundary conditions and loadings

3.2.3 Material properties

High-strength concrete with $f_{c,28} = 77$ MPa is adopted, and 18 key parameters are selected for sensitivity analysis [**liu_stochastic_2001**].

3.3 Implementation of the thermal model

3.3.1 Stationary

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3.3.2 Transient

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3.4 Implementation of the hydric model

3.4.1 Drying

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3.4.2 Shrinkage

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3.5 Implementation of the mechanical model

3.5.1 Elastic

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3.5.2 Mazars

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3.6 Implementation of the TH and THM coupled models

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3.7 Numerical results

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Chapter 4

Model validation

4.1 Experimental data

4.1.1 Experimental tests and material properties

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4.1.2 Thermal, hydraulic and mechanic loadings

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4.1.3 Measured data

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4.2 Validation

4.2.1 Comparison between numerical and experimental results

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4.2.2 Discussions on model performance

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Chapter 5

Probabilistic sensitivity analysis

This chapter follows the foundational framework of Baroth This chapter follows the foundational framework of Baroth [**baroth_introduction_2011**].

5.1 Sources of uncertainty and concrete heterogeneity

Main ideas:

- Concrete is not uniform, so its properties vary from place to place.
- Material parameters (e.g., stiffness and permeability) are uncertain.
- Measurements and model assumptions also add uncertainty.

5.1.1 Motivation for probabilistic assessment

In civil engineering, a structure or mechanical system is considered safe when it remains fit for duty for the duration of its service life, without causing damage to itself or its environment. Traditionally, deterministic models have been used to evaluate structural safety. However, the inevitable gap between a real physical system and a simplified numerical model, along with the unforeseeable nature of future loads and environmental conditions, inherently leads to uncertainty.

To properly manage these risks and optimize the total lifecycle cost, modern engineering codes—such as the Eurocodes—rely heavily on probabilistic and statistical tools. The primary motivation for probabilistic assessment lies in its ability to explicitly quantify the safety of a structure through its probability of failure (P_f) and, conversely, its overall reliability ($R = 1 - P_f$). By adopting a probabilistic format, it becomes much easier to introduce random variables into mechanical behavior models, propagate them through structural analyses, and eventually make objective decisions based on both the probability of an event and the gravity of its consequences.

5.1.2 Classification of uncertainties

Predicting the exact lifetime or mechanical response of a construction project is impossible without accounting for multiple sources of uncertainty. According to reliability theory, these uncertainties can be broadly classified into two main categories: Aleatoric and Epistemic.

- **Aleatoric (inherent or natural) uncertainties:** These uncertainties arise from the natural randomness of the environment and the spatial or temporal variability of material properties. Because they represent the inherent physical variation in nature, aleatoric uncertainties cannot be reduced or eliminated; they can only be accurately quantified (through data acquisition) and taken into account during the design process.
- **Epistemic (knowledge-based) uncertainties:** These stem from a lack of complete knowledge or insufficient data. They include *model uncertainties* (the discrepancy between empirical or theoretical mathematical models and actual physical processes) and *statistical uncertainties* (arising from limited data sampling or poor data collection methods). Furthermore, observation errors heavily contribute to epistemic uncertainty. These errors include measurement inaccuracies (caused by operators or poorly calibrated devices), representativity issues (e.g., deducing stress or rigidity indirectly from strain measurements), and time discrepancies (e.g., changes in material properties between the time of measurement and actual casting). Unlike aleatoric uncertainties, epistemic uncertainties can be effectively reduced by improving physical models, increasing training, and gathering more accurate data.

5.1.3 Concrete heterogeneity and material variability

A critical source of aleatoric uncertainty in civil engineering is the inherent heterogeneity of construction materials, such as concrete and geomaterials.

- **Historical and environmental dependence:** Concrete is not perfectly uniform. Its physical and mechanical properties depend heavily on its entire formation history, the conditions during casting, internal chemical reactions (like alkali-silica reactions), and long-term degradation processes caused by the surrounding environment.

- **Continuous and discrete spatial variations:** Because of this historical dependence, material parameters vary significantly from place to place. This spatial heterogeneity manifests in two ways:

  1. *Continuous variations:* Gradual spatial fluctuations in properties such as stiffness, Young's modulus, Poisson's ratio, density, or friction.
  2. *Discrete variations:* The localized appearances of defects such as micro-cracks, water inlets, networks of capillaries, or weak points within the structure.

Because these material parameters are fundamentally variable, evaluating the true safety and performance of a structure requires geotechnical and structural predictions to abandon purely deterministic values. Instead, they must rely on random variables and stochastic fields to accurately model this spatial heterogeneity.

5.2 Probabilistic framework and implementation

Main ideas:

- Treat uncertain material parameters as random variables.
- Generate many random samples.
- Run the THM model for each sample.
- Collect the outputs and analyze variability and sensitivity.

5.2.1 Implementation: Lognormal - Gaussian transformation

In practical finite element implementations using the Cast3M software, material parameters like the Young's modulus (E) are often assigned a lognormal random field to ensure the generated stiffness values remain strictly positive. However, the built-in random field operator (`ALEA`) generates Gaussian (normal) distributions, necessitating a mathematical transformation.

Assuming the actual mean of the material's Young's modulus is μ_E and its standard deviation is σ_E. The fundamental relationship between these two variables is expressed through the exponential transformation:

$$E = \exp(G) \iff G = \ln(E)$$

From statistical properties, we can establish a system of two equations relating the moments of E (μ_E, σ_E) and G (μ_G, σ_G) as follows:

$$\begin{cases} \mu_E = \exp\left(\mu_G + \dfrac{\sigma_G^2}{2}\right) \\ \sigma_E^2 = [\exp(\sigma_G^2) - 1]\exp(2\mu_G + \sigma_G^2) \end{cases} \tag{5.1}$$

To find the input parameters μ_G and σ_G for the random number generation algorithm, we need to solve the above system of equations.

1. Solving for σ_G^2

By squaring the first equation, we get:

$$\mu_E^2 = \exp(2\mu_G + \sigma_G^2)$$

Substituting this result into the second equation yields:

$$\sigma_E^2 = \left[\exp(\sigma_G^2) - 1\right]\mu_E^2$$

Rearranging the terms, we obtain:

$$\exp(\sigma_G^2) = \frac{\sigma_E^2}{\mu_E^2} + 1$$

Taking the natural logarithm of both sides gives the variance of the Gaussian distribution:

$$\sigma_G^2 = \ln\left(1 + \frac{\sigma_E^2}{\mu_E^2}\right)$$

2. Solving for μ_G

Using the first equation and isolating $\exp(\mu_G)$, we have:

$$\mu_E = \exp(\mu_G) \cdot \exp\left(\frac{\sigma_G^2}{2}\right)$$

Substituting $\sigma_G^2 = \ln\left(1 + \frac{\sigma_E^2}{\mu_E^2}\right)$ into the exponential term, we get:

$$\mu_E = \exp(\mu_G) \cdot \exp\left(\ln\left(\sqrt{1 + \frac{\sigma_E^2}{\mu_E^2}}\right)\right) = \exp(\mu_G) \cdot \sqrt{1 + \frac{\sigma_E^2}{\mu_E^2}}$$

Rearranging to solve for μ_G:

$$\exp(\mu_G) = \frac{\mu_E}{\sqrt{1 + \frac{\sigma_E^2}{\mu_E^2}}} \implies \mu_G = \ln\left(\frac{\mu_E}{\sqrt{1 + \frac{\sigma_E^2}{\mu_E^2}}}\right)$$

Summary of calculation steps in the numerical algorithm

Based on the exact solutions derived above, the implementation follows these sequential steps:

1. The coefficient of variation (cov_E) of the desired lognormal distribution is calculated:

$$cov_E = \frac{\sigma_E}{\mu_E}$$

2. The mean of the equivalent Gaussian distribution (μ_G):

$$\mu_G = \ln\left(\frac{\mu_E}{\sqrt{1 + cov_E^2}}\right)$$

3. The variance of the equivalent Gaussian distribution (σ_G^2):

$$\sigma_G^2 = \ln(1 + cov_E^2)$$

4. The standard deviation of the Gaussian distribution (σ_G):

$$\sigma_G = \sqrt{\ln(1 + cov_E^2)}$$

5. A standard Gaussian random field (`GFIELD`) is then generated using the `ALEA` operator with the computed Gaussian mean μ_G and standard deviation σ_G.

6. Finally, this Gaussian field is transformed back into the actual lognormal distribution (`EFIELD`) using the exponential function. A scale factor of 10^9 is multiplied to convert the physical unit from GPa to Pa for the mechanical solver:

$$EFIELD = \exp(GFIELD) \times 10^9$$

Spatial correlation length (λ)

A fundamental parameter in the generation of random fields is the spatial correlation length, often denoted as λ (or b). While the mean and standard deviation define the statistical distribution of the values at any given single point, the correlation length dictates the spatial structure and the degree of spatial dependence between neighboring points.

Physically, in a heterogeneous material like concrete, the macroscopic properties (such as Young's modulus or permeability) at two closely located spatial points are not entirely independent. They are correlated due to the presence of meso-scale structures, such as aggregates and the cement paste matrix. The correlation length λ mathematically represents the distance over which these macroscopic properties remain statistically correlated.

Depending on the assigned value of λ:

- A very small correlation length ($\lambda \to 0$) implies that the material properties fluctuate rapidly and erratically over short distances, approaching a completely uncorrelated "white noise" spatial distribution.
- A large correlation length ($\lambda \to \infty$) implies a slow spatial variation, moving towards a uniformly homogeneous material where the property is identical across the entire macro-scale structure.

In the context of this study, the choice of the correlation length is closely related to the maximum aggregate size of the concrete mixture. In the finite element implementation using the Cast3M code, this spatial correlation is explicitly introduced into the generation of the log-normal field via the `ALEA` operator using the `'LAMBDA'` keyword. For instance, an exponential covariance function with a specific correlation length b is generated using the Turning Bands Method (`'BANDES_TOURNANTES'` algorithm).

The accurate selection of this parameter is crucial, as it directly governs the tortuosity of the damage patterns and influences the localization of cracks during the high-temperature spalling process.

5.2.2 Iterative Framework and Monte Carlo Simulations

After successfully implementing the log-normal random fields to represent the concrete's spatial heterogeneity, the next step in the probabilistic assessment is to perform multiple independent realizations. This is achieved through a Monte Carlo simulation framework.

Given the inherent complexity and computational cost of the non-linear Thermo-Hydro-Mechanical (THM) model, solving the problem for hundreds of iterations requires a robust automated procedure. In the finite element code Cast3M, this iterative process is seamlessly governed by the `REPETER` operator.

Simulation loop and dynamic material updating

For a predefined number of simulations N_{simu}, the global algorithm performs the following sequential steps within each loop:

1. **Random Field Generation:** A new statistically independent realization of the random field is generated using the `ALEA` operator. For instance, the spatial distribu-

tion of the Young's modulus $\mathbf{E}(x)$ is recalculated based on the assigned log-normal parameters (mean, standard deviation, and correlation length).

2. **Material Updating:** The macroscopic properties of the finite element model are updated. The newly generated random field is projected onto the mesh to form the updated material stiffness matrix using the `MATE` operator.

3. **Non-linear Resolution:** The `PASAPAS` procedure is invoked to solve the multi-physics THM problem from $t = 0$ to the final time t_{fin}, utilizing the updated heterogeneous material properties.

Data extraction and exportation strategy

Storing the complete output database (which includes nodal displacements, stress tensors, and internal damage variables for every time step) for hundreds of simulations would result in an unmanageable amount of data and exceed memory limits. Therefore, a selective data extraction strategy is implemented.

Instead of saving the entire mesh history, a specific "Quantity of Interest" (QoI) is dynamically extracted at the end of each calculation loop. Depending on the targeted sensitivity analysis for the spalling risk, the QoI could be the maximum equivalent plastic strain, the peak pore pressure, or the maximum macroscopic damage variable D.

In the Cast3M script, this is executed using the `EXCO` (to extract the specific component, e.g., the internal variable `'V1'` for damage) and `MAXI` operators to find the extreme value over the spatial domain. Finally, the extracted scalar values are continuously appended to an external structured text file (CSV format) using the `SORT 'CHAI'` command.

This exported dataset containing the input realizations and the corresponding QoI outputs serves as the direct input for the external probabilistic software (e.g., Persalys) to compute the probability density functions and rank the dominant parameters via Sobol indices.

5.3 Sensitivity analysis methodology

Main ideas:

- Sensitivity indicators: Young modulus...
- Change one input parameter at a time (or groups of parameters).
- Measure how much the outputs change.
- Rank the parameters from most to least influential.

5.4 Identification of dominant parameters

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Chapter 6

Conclusions and Perspectives

6.1 Main findings

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6.2 Model limitations and possible improvements

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6.3 Perspectives for future research

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Nomenclature

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